

RH20T: 面向一次性学习多样化技能的综合机器人数据集

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摘要——开放领域中机器人操作面临的一个关键挑战是如何为机器人获取多样化且可泛化的技能。近期一次性模仿学习和机器人基础模型的进展表明，基于演示将训练好的策略迁移到新任务具有广阔前景。这一特性对于使机器人能够习得新技能并提升其操作能力极具吸引力。然而，由于训练数据集的局限性，当前研究界的关注点主要集中在简单场景上，例如推或抓放任务，且仅依赖视觉引导。实际上，存在许多复杂技能，其中一些甚至需要同时借助视觉和触觉感知才能解决。本文旨在释放智能体通过多模态感知泛化至数百种真实世界技能的潜力。为此，我们收集了一个数据集，包含超过 110,000 条接触丰富的机器人操作序列，涵盖多样化的技能、场景、机器人和相机视角，全部采集自真实世界。数据集中的每条序列均包含视觉、力、音频和动作信息。此外，我们还为每条机器人序列提供了相应的人类演示视频和语言描述。我们在校准所有传感器并确保数据集高质量方面投入了大量精力。该数据集已在我们的网站上公开：rh20t.github.io。

单独使用 [21]。这表明在开放环境中学习各种操作时，应考虑更多的感官信息。为了解决这些问题，我们重新审视了机器人操作的数据收集过程。在大多数模仿学习文献中，专家机器人轨迹是通过简化的用户界面手动收集的，例如 3D 鼠标、键盘或虚拟现实遥控器。然而，当机器人与环境进行丰富接触交互时，这些控制方法效率低下且存在安全风险。主要原因是使用三维鼠标或键盘控制不够直观，以及使用虚拟现实遥控器时运动漂移导致的不准确性。此外，没有力反馈的远程操作会降低人类的操作效率。在本文中，我们为机器人配备了力扭矩传感器，并采用了具有力渲染功能的触觉设备，以实现精确高效的数据收集。为了使数据集具有代表性、泛化性、多样性并接近现实，我们收集了约 150 项技能，这些技能包含除简单抓取放置之外的复杂动作。这些技能要么选自 RLBench[19] 和 MetaWorld[40]，要么由我们自己提出。许多技能要求机器人与环境进行丰富接触交互，例如切割、插拔、切片、倾倒、折叠、旋转等。我们使用了全球实验室中常见的多种不同机器人手臂来收集我们的数据集。机器人配置的多样性也有助于算法泛化到其他机器人。

一、引言

机器人操作要求机器人控制其执行器，并按照任务规范改变环境。使机器人能够以最小的努力学习新技能，是机器人学习领域的终极目标之一。最近在单样本模仿学习和新兴基础模型方面的研究，描绘了一幅令人振奋的图景：在给定 π 演示的情况下，将训练好的策略 π 迁移到新任务。本文怀有同样的愿景。尽管前景光明，但机器人领域仍然面临许多挑战。大多数研究仅在简单场景中展示了其算法的泛化性，例如 π 在现实世界中推、抓和放置物体。两个主要因素阻碍了该方向对更复杂任务的探索。首先，尽管社区长期以来一直渴望大规模且多样化的机器人操作数据集，但该领域仍缺乏此类数据集 [6]。根本问题源于数据采集的巨大障碍。这些挑战包括 π 配置多样化机器人平台、创建多样化环境以及收集操作轨迹的艰巨任务，这需要大量的努力和资源。其次，大多数方法仅关注视觉引导控制，然而生理学研究表明，数字感知能力受损的人类在视觉引导下难以完成许多日常操作

到目前为止，我们已经收集了约 110,000 个机器人操作序列和 110,000 个对应相同技能的人类演示视频。这相当于机器人操作序列超过 4000 万帧图像，人类演示超过 1000 万帧。每个机器人序列包含来自多个传感器的丰富视觉、触觉、音频和本体感觉信息。数据集经过精心组织，我们相信具有如此多样性和规模的数据集对于 π 未来基础 π 通用技能学习的出现至关重要，正如自然语言处理和计算机视觉社区所见证的有前景的进展一样 [23]。

II. 相关工作

我们简要回顾了机器人操作数据集、零/单次模仿学习以及视觉-力学习方法的相关工作。

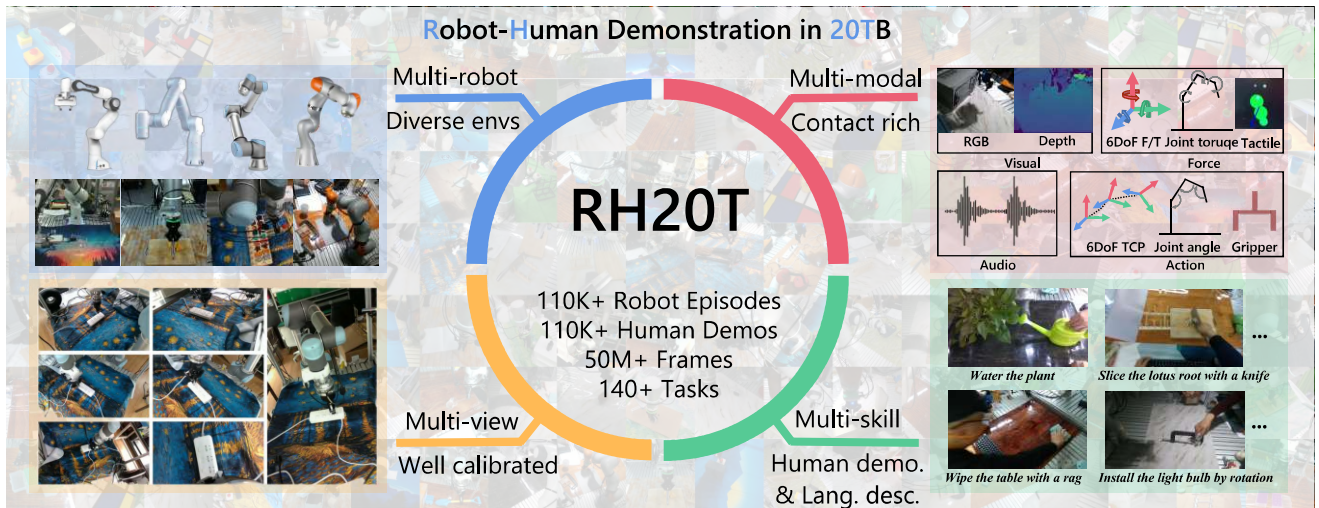


图 1: 概述 our RH20T dataset. 我们采用多个 robots and 设置 diverse environments 用于 the data 采集。robot 操作片段包括多模态视觉、力、音频和动作数据。对于每个片段，我们使用标定良好的多视角相机采集 the 操作过程。我们的数据集包含多样的机器人操作技能，每个片段都有对应的人类演示和语言描述。总计，我们提供超过 11 万个机器人片段和 11 万个对应的人类演示。数据集包含超过 5000 万帧和超过 140 个任务。

a) 数据集: 我们的社区 been 长期以来一直努力 to 创建一个大规模且具有代表性的数据集。先前在单样本模仿学习中的研究要么在真实世界收集机器人操作数据，要么在仿真中收集 (27)。然而，他们的数据集通常规模较小，任务简单。一些尝试已经做出，以创建大规模真实机器人操作数据集 [9] [20] (22) 28 Bal 对于 example, RoboTurk [23] 开发了一个众包平台，并使用基于手机的遥操作在三个任务上收集数据。

[15]

MIME 使 [34] Baxter 通过动作示教收集了 20 种操作，但它们仅限于单一机器人和简单环境。RoboNet a) 收集了大量具有各种机器人、夹爪、and 环境的机器人轨迹。然而，由于执行有意义的 skills. BC-Z [20]

存在挑战，它主要由随机行走片段组成，呈现了一个包含 100 个“任务”的操作集合，但正如 271 中所指出的，它们是 9 个动词和 6-15 个物体的组合。类似地，RT-1 和 RoboSet 21 也收集了大规模操作数据集，但专注于有限的技能集。与我们的工作同时，BridgeData V2 [36] 收集了一个在 24 个环境中包含 13 种技能的数据集。在本文中，我们提出了 [5] 更大的数据集，具有更广泛的技能和环境范围，以及更全面的信息。更重要的是，所有先前的数据集都较少强调接触丰富的操作。我们的数据集更关注这种情况，并包含操作过程中关键的力模式。

b) 零样本/单样本模仿学习: 训练能够基于机器人/人类演示迁移到新任务的策略的目标并不新鲜。早期工作 [B3] 29 专注于使用高层状态 (如轨迹) 进行模仿学习。最近，研究人员 18 El Bi BO #4 5 [35] [4] [39] 8 [26] [20] [27] have started

[14] [10] [42] []

6

探索原始像素输入，借助 the 深度神经网络的进步。此外，通过消除对动作的需求，降低了对示范的要求。近期方法探索了多种一次性任务描述符，包括图像 A]、语言 35 26 20]。这些方法可大致分为三类: 模型无关元学习 [0 4 条件行为克隆 & [20] [3] [21]，and task 图 construction [10] [11]。尽管在该方向上已取得显著进展，但 has 这些 [5] [2]，robot video [14] [8] [27]，or human video [41] 方法仅考虑视觉观测，且主要关注简单的机器人操作，如 as 到达、抓取、推动或放置。我们的数据集提供了进一步发展的机会，使得在单三模仿学习模型内学习数百种需要多模态感知的技能成为可能。

c) Vision 与力的多模态学习: 力感知在操作任务中扮演 a 关键角色，当视觉感知被遮挡时提供有价值且互补的信息。机器人操作中视觉 and 力的联合建模近期引起了研究界的兴趣 [23] T3 24 7 371。然而，这些研究大多忽视了不同模态的异步特性，仅在神经网络之前或之后简单拼接信号。此外，现有研究主要聚焦于为特定任务设计多模态学习算法，如抓取插入 [12]、扭转 [12] 或玩叠叠乐 [13]。A recent 尝试 Ba 探索分别在 6 个任务上联合模仿 the action and wrench。总体而言，如何以连贯方式有效处理不同频率的多模态感知以适应各种技能，这一问题在机器人学中仍未解决。我们的数据集为探索跨多种真实世界技能的多感官学习提供了机遇。

Dataset	# Traj.	# Skills	# Robots	Human Demo	Contact Rich	Depth Sensing	Camera Calib.	Force Sensing
MIME [34]	8.30k	12	1	✓	✗	✓	✗	✗
RoboTurk [28]	2.10k	2	1	✗	✗	✗	✗	✗
RoboNet [9]	162k	N/A	7	✗	✗	✗	✗	✗
BridgeData [11]	7.20k	4	1	✗	✗	✓*	✗	✗
BC-Z [20]	26.0k	3	1	✓	✗	✗	✗	✗
RoboSet [2]	98.5k	12	1	✗	✓	✓	✗	✗
BridgeData V2 [36]	60.1k	13	1	✗	✓	✓*	✗	✗
RH20T	110k	42	4	✓	✓	✓	✓	✓

表一：与以往公开数据集的比较：“相机校准”表示所有相机与机器人的外参校准。表示仅部分图像配有深度感知。这一比较凸显了我们数据集的全面性，它是迄今为止最广泛的机器人操作数据集。

Conf.	Robot	Gripper	6DoF F/T Sensor	Tactile
Cfg 1	Flexiv	Dahuan AG95	OptoForce	N/A
Cfg 2	Flexiv	Dahuan AG95	ATI Axia80-M20	N/A
Cfg 3	UR5	WSG50	ATI Axia80-M20	N/A
Cfg 4	UR5	Robotiq-85	ATI Axia80-M20	N/A
Cfg 5	Franka	Franka	Franka	N/A
Cfg 6	Kuka	Robotiq-85	ATI Axia80-M20	N/A
Cfg 7	Kuka	Robotiq-85	ATI Axia80-M20	uSkin

TABLE II: Hardware specification of different configurations.

Conf.	Modal	Size	Frequency
Cfg 1-7	RGB image	1280×720×3	10 Hz
	Depth image	1280×720	10 Hz
	Binocular IR image	1280×720	10 Hz
	Robot joint angle	6 / 7	10 Hz
	Robot joint torque	6 / 7	10 Hz
	Gripper Cartesian pose	6 / 7	100 Hz
	Gripper width	1	10 Hz
	6DoF F/T	6	100 Hz
	Audio	N/A	30 Hz
Cfg 7	Tactile	2×16×3	200 Hz

表三：不同配置的数据信息。前 9 种数据模式对所有机器人配置均相同。最后一种指尖触觉感知数据模式仅在配置 7 中可用。

三、RH20T 数据集

我们向社区介绍我们的机器人操作数据集，即 20TB 规模的机器人-人类演示数据集 (RH20T)。图 1 展示了我们的数据集的概览。

A. RH20T 的特性

RH20T 的设计目标是实现通用机器人操作，即机器人能够根据任务描述（通常是人类演示视频）执行各种技能，同时尽量减少对刚性任务的依赖。以下特性强调实现这一目标，表 1 提供了我们的数据集与以往具有代表性的公开数据集之间的比较。a) 多样性：RH20T 的多样性涵盖多个方面。为确保任务多样性，我们选择了 RLBench [19] 中的 48 个任务、MetaWorld [EO] 中的 29 个任务，并引入了 70 个自拟任务，这些任务机器人经常遇到且能够完成。总计包含

147 tasks, consisting of 42 skills (*i.e.*, verbs). Hundreds of objects were collected to accomplish these tasks. To ensure applicability across different robot configurations, we used 4 popular robot arms, 4 different robotic grippers, and 3 types of force-torque sensors, resulting in 7 robot configurations. Details about the robot configurations are provided in Tab. II.

To enhance environment diversity, we frequently replaced over 50 table covers with different textures and materials, and introduced irrelevant objects to create distractions. Manipulations were performed by tens of volunteers, ensuring diverse trajectories. To increase state diversity, for each skill, volunteers were asked to change the environmental conditions and repeat the manipulation 10 times, including variations in object instances, locations, and more. Additionally, we conducted robotic manipulation experiments involving human interference, both in adversarial and cooperative settings. Further details about each task are provided in the appendix.

b) *Multi-Modal*: We believe that the future of robotic manipulation lies in multi-modal approaches, particularly in open environments, where data from different sensors will become increasingly accessible with advancements in technology. In the current version of RH20T, we provide visual, tactile, audio, and proprioception information. Visual perception includes RGB, depth, and binocular IR images from three types of cameras. Tactile perception includes 6 DoF force-torque measurements at the robot’s wrist, and some sequences also include fingertip tactile information. Audio data includes recordings from both in-hand and global sources. Proprioception encompasses joint angles/torques, end-effector Cartesian pose and gripper states. All information is collected at the highest frequency supported by our workstation and saved with corresponding timestamps, and the details are given in Tab. III.

c) *Scale*: Our dataset consists of over 110,000 robot sequences and an equal number of human sequences, with more than 50 million images collected in total. On average, each skill contains approximately 750 robot manipulations. Fig. 2 provides a detailed breakdown of the number of manipulations across different tasks in the dataset, showing a relatively uniform distribution. Fig. 3 presents statistics on the manipulation time for each sequence in our dataset. Most sequences have durations ranging from 10 to 100 seconds. With its substantial volume of data, our dataset stands as the

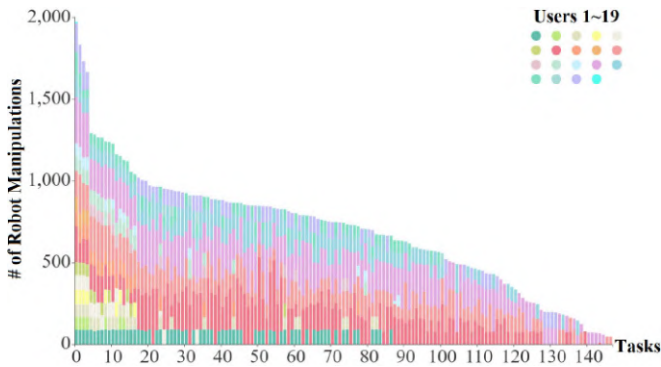


图 2: 针对 different 个任务的操作统计 on the amount robotic.

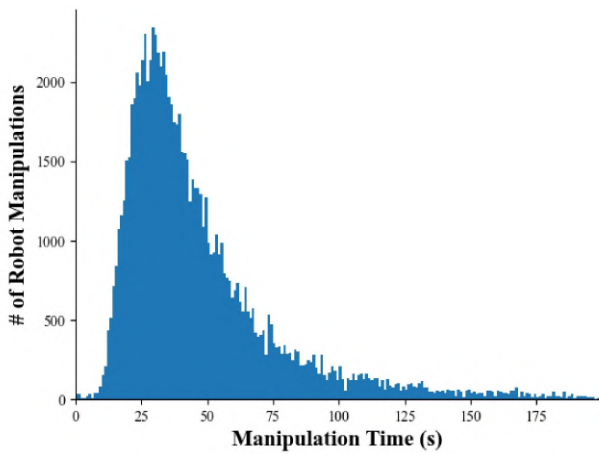


图 3: 我们数据集中操作 different robotic 的统计 on the execution time.

目前我们社区中规模最大的。

da) 数据层级: 人类能够基于视觉观察准确理解任务的语义, 无论 the 视角、背景、操作主体或对象如何。我们旨在提供一个包含密集的<人类演示, 机器人操作>对的数据集, 使模型能够学习这一特性。为实现这一目标, 我们根据任务内相似性以树状层级组织数据集。图展示了一个示例树结构及不同层级的标准。具有更近共同祖先的叶节点关系更密切。对于每个任务, 通过在不同层级上配对具有共同祖先的叶节点, 可以构建数百万个<人类演示, 机器人操作>对。e) 组合性: RH20T 不仅包含执行单一操作的 not 序列 short, 还包含组合多个短任务的长操作序列。例如, 抓取插头、将其插入插座、打开插座开关以及打开台灯等一系列动作可以被视为一个单一任务, 其中每个步骤本身也是一个任务。这种任务组合使我们能够研究掌握短序列是否有助于长序列任务的习得。

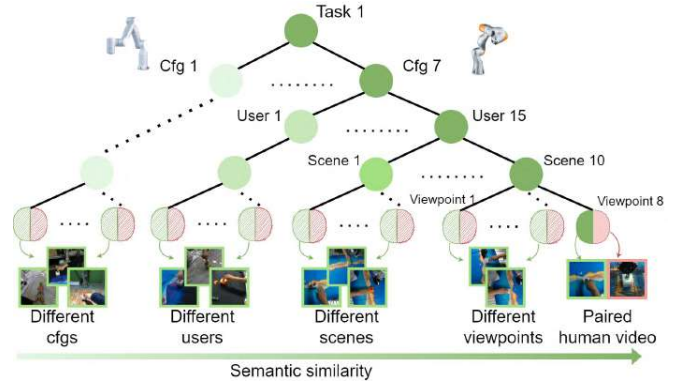


图 4: data 层次结构示例: 叶 nodes in the 层次结构由人类演示 (以绿色高亮) 和机器人操作 (以红色高亮, 图中仅显示最右侧示例) 组成。我们可以将机器人操作序列与从不同视角、场景、人类主体和环境捕获的人类演示视频配对。放大以探索 the 各种人类演示的细节。

B. 数据收集与处理

与以往使用 3D 鼠标、虚拟现实遥控器或手机简化遥操作接口的方法不同, 我们强调 the 直观 and 准确性的重要性在收集接触丰富的机器人操作数据时进行遥操作。没有适当的遥操作, 机器人很容易与环境碰撞并产生显著的力, 触发紧急停止。因此, 以往的工作要么避免接触 [20], 要么以降低的速度运行以减轻这些风险。

a) 采集: 图 2 展示了我们 data 采集平台的一个 an 示例。每个平台包含一个配备力扭矩传感器、夹爪和 1-2 个手部相机的机械臂, 8-10 个全局相机, 2 个麦克风, 一个触觉设备, 一个踏板以及一个数据采集工作站。所有相机在执行操作前均已完成外参标定。人类演示视频在同一平台上由人类佩戴额外的第一视角相机采集。数十名志愿者根据我们的任务列表和文本描述执行机器人操作。我们使遥操作非常直观, 平均培训时间少于 1 小时。志愿者还需在每次操作完成后指定任务结束时间, 并给出 0 到 9 的评分。0 表示机器人进入紧急状态 (例如硬碰撞), 1 表示任务失败, 2-9 表示他们对操作质量的评估。在我们的数据集中, 成功与失败案例的比例约为 10:1。b) 处理: 我们对数据集进行预处理, 以提供一致的数据接口。所有机器人和力扭矩传感器的坐标系均已对齐。不同的力扭矩传感器经过仔细去皮。末端执行器笛卡尔位姿和力扭矩数据被转换到每个相机的坐标系中。对每个场景进行人工验证, 以确保相机标定质量。图展示了在统一坐标系中渲染数据不同组件的示意图, 并演示了

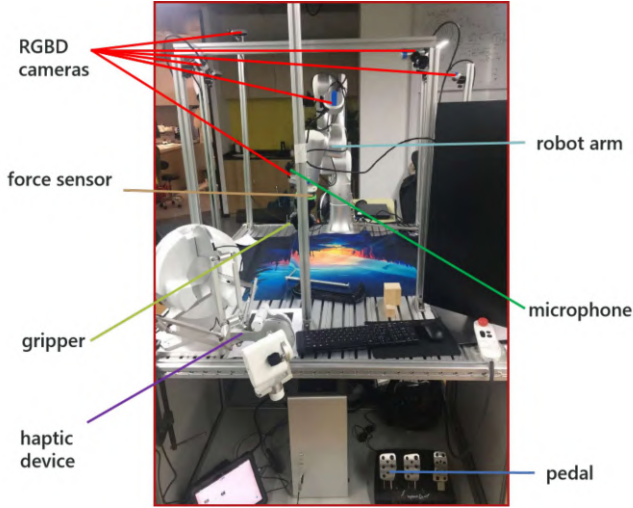


图 5: Illustration of our data 采集平台

我们数据集的高质量。详细的 data 格式 and 数据访问应用程序编程接口在我们的网站上提供。

四、实验

我们引入 RH20T 数据集, 旨在使机器人能够在陌生环境中利用最少数据习得新技能。尽管最终目标是训练一个能够以小样本学习方式执行此类任务的大型模型, 但我们承认这一努力所需的巨大计算资源目前超出我们的能力范围。因此, 本文主要聚焦于展示该数据集在提升基线模型于小样本学习框架内的可迁移性方面的有效性。为评估我们数据集的有效性, 我们采用动作

Chunking with Transformers (ACT) model as our baseline 网络。ACT 由近期工作 [M3] 提出, 在处理复杂机器人操作任务方面展现出卓越能力。它利用变换器的力量从数百次演示中学习复杂的动作序列。

A. 实验设置

- 平台: 在我们的实验中, 我们使用一台 Flexiv 机械臂, 配备位于机器人前方的英特尔实感 RGB-D 相机以感知环境, 以及一个大寰-95 夹爪用于与物体交互。我们搭建了一个新环境, 其中相机位姿和桌面覆盖物与我们的 RH20T 数据集中的不同。图 7 (a) 展示了我们的机器人平台。
- 流程: 我们设置的任务包括抓取一个方块并将其放置在一个重物上。在新环境中, 我们收集了 75 个机器人操作序列, 包括 RGB 图像与动作, 通过遥操作采集。从我们的数据集中, 我们选取了 335 个来自同一任务的机器人操作序列, 以及 195 个来自 3 个不同但相似任务的操作序列 (抓取

up a 方块; 抓取 up a 方块并 place it at the 指定位置, 抓取 up a 方块 and move it from left to 右侧)。所有



图 6: We 展示了通过融合 the 来自我们数据采集平台上安装的多视角摄像头的 RGBD 数据而生成的 the point cloud。红色金字塔表示相机位姿。此外, 机器人模型根据数据集中记录的关节角度在场景中渲染。显然, 所有相机都相对于机器人基座坐标系进行了标定, 并且所有记录的数据在时间域上是同步的。

我们数据集中的操作序列与当前实验中机器人环境具有不同的相机视角、桌面覆盖物、物体和机器人本体。

我们通过在数据集中选取的不同数据子集上预训练 ACT 模型来启动训练过程。通过让模型接触 a range 种机器人操作场景, 我们旨在增强其跨各种任务和环境条件的泛化能力。预训练之后, we fine-tune the ACT model on 新收集数据中的特定部分, 专注于涉及抓取和放置重物的任务。此阶段旨在 to 细化 the 模型在目标任务上的性能。

我们评估了 ACT 模型在有和没有在我们的数据集上预训练两种情况下的性能。实验在真实机器人平台上进行, 每种配置重复 20 次。我们将任务分为 3 个阶段, 即机器人是否能够到达方块、抓取它并将其放置在重物上, 并测量每个阶段的成功率。此外, 我们考察了模型对 to 物体属性变化的泛化能力。评估时间限制设置为 60 秒。c) 实现 Details: 对于 ACT 模型, we set the 网络中的隐藏通道和前馈通道分别设置为 512 和 3200。在预训练阶段, 模型以 2×10^{-5} 的学习率训练 10 个训练轮次; 而在微调阶段, 模型以 10^{-3} 的学习率训练 750 个训练轮次。尽管这少于原始实现 [3], 但我们通过包含新收集演示的所有有效子轨迹来增加每个训练轮次的样本密度。因此,

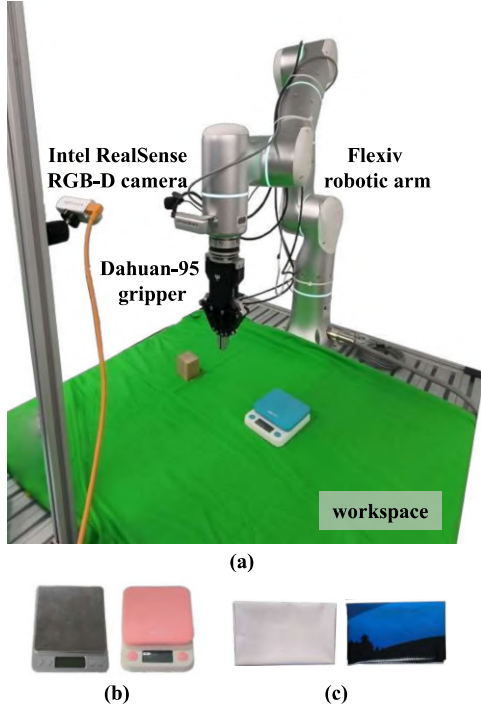


图 7: (a) 实验机器人平台。(b) 不同重物 (金属、粉色) 评估 the 模型的泛化能力。(c) 不同桌布 (白色、蓝色) 评估模型的泛化能力。

750 epochs are sufficient for the model to converge well. The chunk size is set to 20, which corresponds to 2 seconds with the frequency of 10Hz. The images are scaled to 640×360 during training and testing. We apply temporal ensembling and set its coefficient $k = 0.01$ following [43] in evaluation.

B. 实验结果

We present the model's success rates under different 表 中的训练配置。当使用 75 个演示训练网络时，我们观察到，尽管相机视角、机器人本体和背景存在差异，使用我们数据集中选定的数据对模型进行预训练，仍能提高最终成功率。此外，在预训练期间加入来自不同任务的数据进一步提升了 the 整体成功率 rate.。将训练 500 个训练轮次并进行预训练的结果与训练 750 个训练轮次但不进行预训练的结果进行比较 the，我们发现使用我们的数据集进行预训练也加速了模型收敛。这些结果表明，利用我们数据集中多样化的训练数据增强了机器人操作模型的适应性和鲁棒性。

随后，我们减少在该新环境中收集的演示数量，以模拟小样本学习场景。在仅使用 40 个机器人演示的情况下，基于我们数据集预训练的结果优于未预训练且使用 75 个演示训练的对照模型。进一步将演示数量减少至 10 个时，基于我们数据集中多个任务预训练的结果仍超越使用 75 个演示训练的模型。

# Demos	Pretrain Task		Training Epochs	Success Rate (%) ↑		
	Same	Multi.		Reach	Pick	Place
75			500	35	10	0
	✓		500	70	15	15
	✓	✓	500	65	20	15
			750	55	5	0
	✓		750	80	20	15
	✓	✓	750	80	25	25
40			750	45	10	0
	✓		750	65	25	5
	✓	✓	750	70	25	15
10			750	15	0	0
	✓		750	30	15	5
	✓	✓	750	50	10	5

表 4: ACT trained in different 的实验 results 在原始环境中进行设置与测试 (20 次试验)。

Type	w/wo pretrain	Success Rate (%) ↑		
		Reach	Pick	Place
weight	metal	20	0	0
	✓	80	30	10
pink		40	10	0
	✓	70	20	10
table cover	white	20	0	0
	✓	50	0	0
	blue	30	0	0
	✓	80	20	10

表 V: 在不同设置下训练并在不同环境中测试的 ACT 的实验结果 (10 次试验)。

无需预训练的演示。这证明了我们的数据集对机器人操作中样本学习的有益影响。

最后，我们在测试阶段将 the 物体 and 桌布 used 替换为全新实例，以评估模型在新环境中的泛化能力。用于替换的砝码和桌布如图 (b) 和 (c) 所示。本实验比较两个模型，二者均使用原始 75 个演示训练 750 轮次，其中一个在数据集中的多个相似任务上进行了预训练，另一个则没有。表中的实验结果表明，在数据集上预训练的模型始终优于未预训练的模型，说明该数据集增强了模型的泛化能力。

V

五、讨论与结论

本文提出了面向多样化机器人技能学习的 RH20T 数据集。我们相信该数据集能够促进机器人学多个领域的发展，尤其是新环境下机器人操作任务。本文目前的局限性在于：(一) 数据采集成本高昂；(二) 机器人基础模型的潜力尚未在该数据集上得到评估。我们曾尝试复现近期一些机器人基础模型的结果，但由于计算资源限制尚未成功。因此，我们决定现阶段开源该数据集，并希望与社区共同推动该领域的发展。未来，我们

希望将数据集扩展到更广泛的机器人操作任务，包括双臂和多指灵巧操作。

作者贡献 方浩树发起了项目, 搭建了机器人平台, 初始化了遥操作工具包, 整理了数据采集流程, 并撰写了论文。方浩搭建了机器人平台, 优化了遥操作工具包, 协助策略训练, 并撰写了 the 项目页面。Z. 唐协助数据采集, 校准传感器, 构建数据集, 并编写了数据访问应用程序编程接口。王辰训练了策略。刘军探索了基于变换器架构的单样本模仿学习。王杰协助数据采集和数据集解析。朱浩探索了为人类演示视频标注人体关键点。卢策吾指导了项目并提供硬件和资源支持。

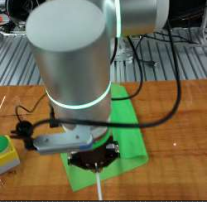
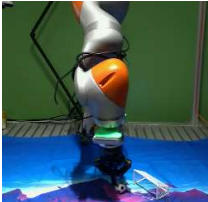
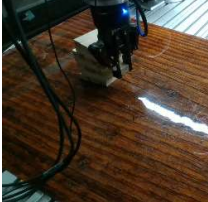
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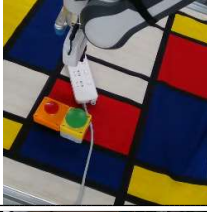
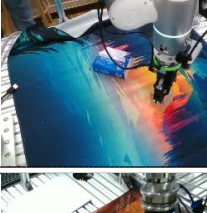
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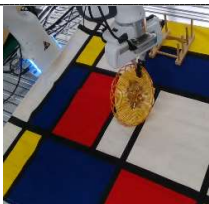
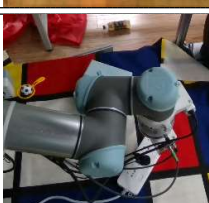
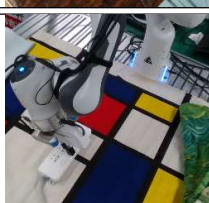
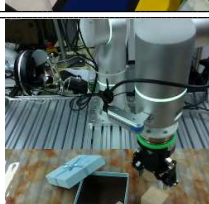
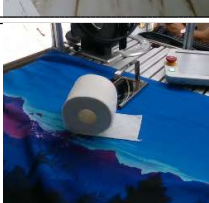
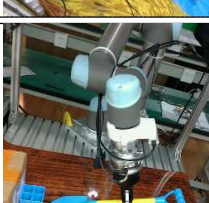
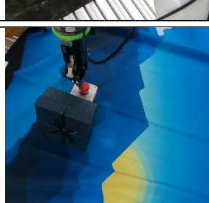
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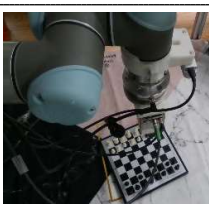
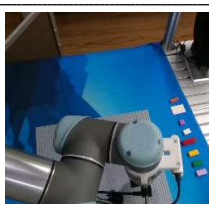
附录 RH20T 任务规范

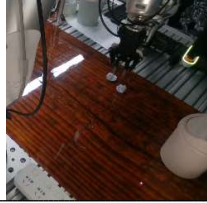
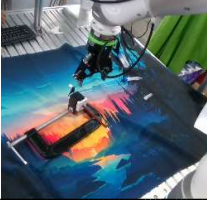
表 1: 数据集任务描述。“来源”表示任务来源。注意任务编号不一定连续。

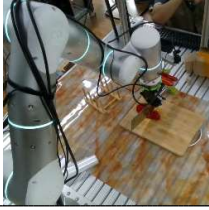
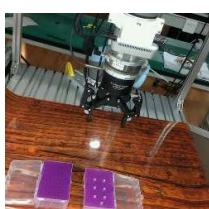
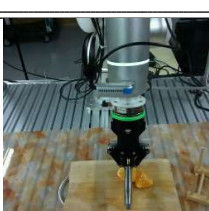
Items	Task Desc.	Src.	Items	Task Desc.	Src.	Items	Task Desc.	Src.
	1. Press the button from top to bottom	Meta-World		2. Pull out a napkin	Self-Proposed		3. Press three buttons from left to right in sequence	RLBench
	4. Pick up a block on the left and move it to the right	Meta-World		5. Approach and touch the side of a block	Meta-World		6. Use the gripper to push a block from left to right	Meta-World
	7. Hold a block with the gripper and sweep it from left to right on the table	Meta-World		8. Grab a block and place it at the designated location	RLBench		9. Take out one Hanoi block and throw it aside	RLBench
	10. Place the handset of the telephone on the corresponding phone cradle	RLBench		11. Water the plant	RLBench		12. Push the soccer ball into the goal	Meta-World
	13. Place the block on the table 缩放	RLBench		14. 重新 - move the object from the 缩放	RLBench		15. Play the drum	Self-Proposed
	16. 击打 pool ball	RLBench		17. 放置 the pen into the 笔筒	RLBench		18. Play Jenga	RLBench

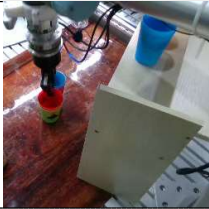
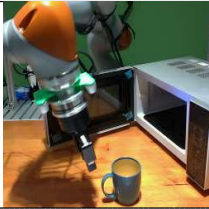
Items	Task Desc.	Src.	Items	Task Desc.	Src.	Items	Task Desc.	Src.
	19. Play the first move as black in the upper right corner of the Go board	Self-Proposed		20. Turn on the desk lamp by pressing the button	RLBench		21. Turn off the desk lamp by pressing the button	RLBench
	22. Wave the flag	Self-Proposed		23. Turn on the power strip by pressing the button	Self-Proposed		24. Turn off the power strip by pressing the button	Self-Proposed
	25. Unfold a piece of paper	Self-Proposed		26. Use the gripper to push and close the drawer	Meta-World		28. Grasp the handle and close the drawer	RLBench
	29. Grasp the handle and open the drawer	RLBench		30. Pour out the test tube	Self-Proposed		31. Cover the box	Meta-World
	32. Slide the outer casing onto the gift box	Self-Proposed		33. Grasp one block to sweep the other block onto the mark	Meta-World		34. Stack the squares into a pyramid shape	RLBench
	35. Pick up one small block	RLBench		36. Shake the test tube	Self-Proposed		37. Stack the blocks in a vertical line of five	RLBench
	38. Pick up the cup	RLBench		39. Pour the water from one cup into another empty cup	blabla		40. Stack the cups	RLBench
	41. Clean the table-top with a sponge	RLBench		42. Screw the lid onto the jar	RLBench		43. Unscrew the lid from the jar	RLBench

Items	Task Desc.	Src.	Items	Task Desc.	Src.	Items	Task Desc.	Src.
	44. Pick up a bag of things	Self-Proposed		45. Hang the brush on the pen rack	Self-Proposed		46. Hang the cup on the cup rack	RLBench
	47. Take the cup off the cup rack	RLBench		48. Rotate the steering wheel 90 degrees clockwise	Self-Proposed		49. 旋转 the steering wheel 90 degrees counter-clockwise	Self-Proposed
	50. Put the dish on the dish rack	Self-Proposed		51. Take the dish off the dish rack	Self-Proposed		52. 抓取 a basketball, release it and shoot it into the basket	Meta-World
	53. Use a clamp	Meta-World		54. Catch the moving object	Self-Proposed		55. Transfer liquid using a dropper	Self-Proposed
	56. Receive something handed over by a human	Self-Proposed		57. Turn on the four buttons on the power strip	Self-Proposed		58. Turn off the four buttons on the power strip	Self-Proposed
	59. Turn the knob to increase the volume of a speaker	Self-Proposed		60. Turn the knob to decrease the volume of the speaker	Self-Proposed		61. Take everything out of the gift box	Self-Proposed
	62. Put the toilet paper on its holder	Self-Proposed		63. Use a shovel to scoop up an object	Self-Proposed		64. Take the toilet paper off its holder	Self-Proposed
	65. Build with small Lego blocks	Self-Proposed		66. Build with large Megabloks	Self-Proposed		67. Press a button from top to bottom with obstacles	Meta-World

Items	Task Desc.	Src.	Items	Task Desc.	Src.	Items	Task Desc.	Src.
	68. Press a button horizontally with obstacles	Meta-World		69. Assemble one piece of a puzzle	RLBench		70. Open a sliding window	Meta-World
	71. Close a sliding window	Meta-World		72. Drop coins into a piggy bank	Self-Proposed		73. Put things in the drawer	RLBench
	74. Press the button horizontally	Meta-World		75. Finish setting up the starting position of a chess-board that is almost arranged	Self-Proposed		76. Stack blocks (small Lego) one on top of the other every time	Self-Proposed
	77. Stack blocks (small Lego) randomly one at a time	Self-Proposed		78. Close the microwave door	RLBench		79. Open the microwave door	RLBench
	80. Flip over and spread out the paper that is laid flat on the table	Self-Proposed		81. Unfold the leg of the glasses (with one hand)	Self-Proposed		82. Scoop water with a large spoon from one bowl to another	Self-Proposed
	83. Swat with a flyswatter	Self-Proposed		84. Assemble: Attach the bubble ring to the ball	Meta-World		85. Remove the bubble ring from the assembled bubble ring and ball	Meta-World
	86. Dial a number on an old rotary phone	Meta-World		88. Pick up and place an object with obstacles	Meta-World		89. Push an object with obstacles	Meta-World
	90. Approach and touch an object with obstacles	Meta-World		91. Move an object from one box to another	Meta-World		92. Turn the hands of a clock	RLBench

Items	Task Desc.	Src.	Items	Task Desc.	Src.	Items	Task Desc.	Src.
	93. Put the photo frame on the bracket	RLBench		94. Open a box	RLBench		95. Take the photo frame down from the bracket	RLBench
	96. Take something out of a drawer	RLBench		100. Stir the beaker with a glass rod	Self-Proposed		101. Clean the table with a cloth	Self-Proposed
	102. Scrub the table with a brush	Self-Proposed		103. Drag the plate to the goal post after holding it down	Meta-World		104. Drag the plate back after holding it down	Meta-World
	105. Put the object on the shelf	Meta-World		106. Take the object down from the shelf	Meta-World		107. Put the garbage in the trash can	RLBench
	108. Sharpen the pencil with a pencil sharpener	Self-Proposed		109. Insert the pencil into the pencil sharpener	Self-Proposed		110. Take the pencil out from the pencil sharpener	Self-Proposed
	111. Put the object with the corresponding shape into the corresponding hole	RLBench		112. Plug in the charger to the socket	Self-Proposed		116. Use the correction tape on paper	Self-Proposed
	118. Turn on the water tap	Meta-World		119. Turn off the water tap	Meta-World		120. Install the light bulb by rotating it	RLBench
	121. Take out the light bulb by rotating it	RLBench		122. Put the knife on the cutting board	RLBench		123. Put the knife on the knife rack	RLBench

Items	Task Desc.	Src.	Items	Task Desc.	Src.	Items	Task Desc.	Src.
	124. Push down the lever	Meta-World		125. Pull up the lever	Meta-World		126. Plug in the power cord to the socket	Self-Proposed
	127. Plug in the power cord of the desk lamp, turn on the socket, and light up the desk lamp	Self-Proposed		128. Plug in the USB drive to the docking station	RLBench		129. Plug in the bulb holder with a bulb to the socket	Self-Proposed
	130. Plug in the bulb holder with a bulb to the socket and turn on the switch of the bulb	Self-Proposed		131. Stack the blocks into a pyramid	Self-Proposed		132. Stack the blocks into a cross shape	Self-Proposed
	200. Insert the tip of a large pipette into the holder for large pipette tips	Self-Proposed		201. Insert the tip of a medium pipette into the holder for medium pipette tips	Self-Proposed		202. Insert the tip of a small pipette into the holder for small pipette tips	Self-Proposed
	204. Transfer all large pipette tips from one holder to another holder for large pipette tips	Self-Proposed		205. Chop the scallions	Self-Proposed		206. Chop the green garlic	Self-Proposed
	207. Chop the chili peppers	Self-Proposed		208. Slice the lotus root	Self-Proposed		209. Slice the carrots	Self-Proposed
	210. Chop the onions	Self-Proposed		211. 移液-fer all medium pipette tips from one rack to another holder for medium pipette tips	Self-Proposed		212. Transfer all small 移液管 tips from one rack to another holder for small pipette tips	Self-Proposed
	213. Chop the orange	Self-Proposed		215. Chop the potatoes	Self-Proposed		216. Chop the cucumber into 碎片	Self-Proposed

Items	Task Desc.	Src.	Items	Task Desc.	Src.	Items	Task Desc.	Src.
	217. Plug in the bulb holder to the socket	Self-Proposed		218. Plug in the bulb holder to the socket, install the bulb, turn on the socket to light up the bulb	Self-Proposed		222. Cover the pot with the lid	RLBench
	223. Take the cups off the shelf and stack them together	Self-Proposed		225. Put the bowl into the microwave	Self-Proposed		329. Put the glass cup onto the shelf	Self-Proposed